Cloud & Infrastructure / DevOps Training

Course Syllabus — Module-Based Structure

Module No.	Title
Module 1	Introduction to Cloud Computing
Module 2	Virtualization and Cloud Infrastructure
Module 3	Cloud Service Providers Overview
Module 4	Cloud Networking
Module 5	Cloud Security and Compliance
Module 6	Cloud DevOps and Automation
Module 7	Multi-Cloud and Hybrid Cloud Strategies
Module 8	Cloud Cost Optimization and Performance Management
Module 9	Cloud Computing Real-World Applications
Module 10	Cloud Certification Preparation & Career Readiness

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Course Overview

The **Cloud Computing Certification Course** is designed to provide learners with a complete and structured understanding of modern cloud computing technologies. The course begins with the **fundamentals of cloud computing** and gradually progresses toward advanced topics such as cloud infrastructure, networking, security, DevOps automation, multi-cloud architecture, cost optimization, and real-world cloud applications. Learners will gain exposure to major cloud platforms, including **Amazon Web Services (AWS), Microsoft Azure, and Google Cloud Platform (GCP)**, while developing practical skills through hands-on exercises and projects.

The course focuses on both **theoretical knowledge and practical implementation**. Learners will understand how cloud infrastructure is designed, deployed, secured, monitored, and optimized in real-world environments. Practical activities will help learners work with cloud services such as compute, storage, networking, identity management, containers, Infrastructure as Code, and CI/CD pipelines. The course also introduces real-world cloud architectures and industry use cases to help learners understand how cloud technologies are applied to business and technical challenges.

Duration

The course consists of **10 comprehensive modules** with an estimated learning duration of **60–80 hours**. The duration may vary depending on whether the course is delivered through instructor-led training or self-paced learning. Each module includes theoretical concepts, practical exercises, activities, case studies, and assessments to reinforce learning.

Format

The course follows a combination of **lectures, hands-on labs, case studies, assignments, and a capstone project**. Lectures provide the conceptual foundation, while hands-on labs allow learners to implement cloud services and technologies in practical environments. Case studies demonstrate how cloud solutions are applied in real-world scenarios. The capstone project brings together the concepts learned throughout the course into an end-to-end cloud solution.

Prerequisites

No advanced technical background is mandatory to begin the course. However, a **basic understanding of computer networks and operating systems** is helpful for understanding topics such as IP addressing, virtual machines, servers, networking, and

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cloud infrastructure. Learners with basic programming, Linux, or IT knowledge may find it easier to progress through the practical components, but the course is structured to introduce the required concepts progressively.

Target Audience

This course is suitable for a wide range of learners who are interested in building careers in cloud computing and related technologies. It is particularly suitable for **IT professionals, system administrators, developers, students, software professionals, and career-changers** who want to develop practical cloud skills.

The course is also beneficial for learners preparing for roles such as **Cloud Administrator, Cloud DevOps Engineer, Cloud Infrastructure Engineer, Cloud Security Engineer, and Cloud Architect**. By combining cloud fundamentals with practical deployment, automation, security, and certification preparation, the course helps learners develop skills that can be demonstrated through projects and technical interviews.

Course Objectives

By the end of the **Cloud Computing Certification Course**, learners will have developed both theoretical knowledge and practical skills required to understand, deploy, manage, secure, automate, and optimize cloud-based environments. The course objectives are designed to progressively build cloud expertise from fundamentals to job-ready capabilities.

1. Explain Core Cloud Computing Concepts, Models, and Service Types

Learners will develop a strong foundation in cloud computing and understand how cloud technology differs from traditional on-premises infrastructure. They will learn the key characteristics of cloud computing, including scalability, elasticity, resource pooling, on-demand access, and cost efficiency. Learners will also understand different cloud deployment models such as **Public Cloud, Private Cloud, Hybrid Cloud, and Multi-Cloud**, along with service models including **IaaS, PaaS, SaaS, and FaaS**. This knowledge will help learners identify the appropriate cloud model and service type for different technical and business requirements.

2. Compare and Work Across AWS, Azure, and GCP

Learners will gain practical exposure to the three major cloud service providers: **Amazon Web Services (AWS), Microsoft Azure, and Google Cloud Platform (GCP)**. They will explore commonly used services for compute, storage, databases, networking, identity management, monitoring, and security. Learners will understand the similarities and differences between equivalent services across platforms and develop the ability to select

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suitable cloud services based on project requirements, scalability, availability, security, and cost considerations.

3. Design and Deploy Cloud Infrastructure Using IaaS, PaaS, SaaS, and FaaS Models

Learners will understand how cloud infrastructure is designed and deployed using different cloud service models. They will gain knowledge of virtual machines, storage, databases, managed platforms, software services, and serverless functions. Through practical exercises, learners will learn how to provision resources, configure services, deploy applications, and understand the responsibilities of cloud providers and customers under each service model.

4. Implement Cloud Networking, Security, and Compliance Best Practices

Learners will develop the ability to design secure cloud environments by understanding **VPCs, subnets, IP addressing, DNS, routing, firewalls, security groups, and load balancers**. They will also learn security practices such as **Identity and Access Management (IAM), authentication, authorization, encryption, data protection, auditing, and logging**. The course introduces major compliance standards such as **GDPR, HIPAA, ISO 27001, and PCI-DSS**, helping learners understand the importance of security, governance, and regulatory requirements in cloud environments.

5. Apply DevOps Automation Using CI/CD, IaC, and Containers

Learners will understand how DevOps practices improve the speed, consistency, and reliability of cloud application delivery. They will learn **Continuous Integration and Continuous Deployment (CI/CD)** concepts and gain exposure to tools such as Git, Jenkins, and cloud-based DevOps platforms. Learners will also explore **Infrastructure as Code (IaC)** using Terraform, Ansible, and CloudFormation, along with container technologies such as **Docker and Kubernetes**. These skills will enable learners to automate infrastructure provisioning, application deployment, configuration management, and containerized workloads.

6. Architect Multi-Cloud and Hybrid Solutions and Optimize Cost and Performance

Learners will understand how organizations use **multi-cloud and hybrid-cloud architectures** to meet requirements related to flexibility, availability, workload distribution, migration, and vendor management. They will learn about cloud-to-cloud integration, API interoperability, workload management, and data migration strategies. The course also develops skills in **cloud cost optimization and performance management**, including resource rightsizing, pricing models, scaling, utilization

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monitoring, and identifying opportunities to reduce unnecessary cloud expenditure while maintaining application performance.

7. Prepare for Industry Certifications and Cloud Career Roles

Learners will receive guidance for pursuing industry-recognized certifications across **AWS, Microsoft Azure, and GCP**. The course will support certification preparation through conceptual revision, practical labs, scenario-based questions, and hands-on exercises. In addition, learners will develop career-ready skills through **capstone projects, cloud portfolio development, resume preparation, and technical interview practice**. The objective is to help learners understand potential career paths such as **Cloud Architect, Cloud DevOps Engineer, Cloud Security Engineer, Cloud Administrator, and Cloud Infrastructure Engineer**.

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Module 1: Introduction to Cloud Computing

Estimated Duration: 5–6 hours

This module establishes the foundation of cloud computing. Learners will understand how cloud platforms work, why organizations adopt cloud infrastructure, and the different models used to deliver cloud services.

1. Definition and Key Characteristics of Cloud Computing Introduces cloud computing as the delivery of computing resources such as servers, storage, databases, networking, and applications over the internet. Learners will understand key characteristics such as on-demand resources, scalability, elasticity, resource pooling, and measured usage.

2. History and Evolution of Cloud Technology Explores the evolution from traditional physical servers and data centers to virtualization, cloud infrastructure, and modern cloud-native architectures. Learners will understand the major developments that led to today's cloud platforms.

3. Benefits of Cloud Computing

- Scalability: Resources can be increased or decreased according to workload requirements.
- Cost Efficiency: Organizations can reduce infrastructure and operational costs versus maintaining physical infrastructure.
- Flexibility: Teams can quickly provision infrastructure and deploy applications from different locations.

4. Challenges and Risks in Cloud Adoption Covers security concerns, data privacy, vendor dependency, service availability, migration complexity, compliance requirements, and unexpected cloud costs.

5. Cloud Deployment Models

- Public Cloud: Infrastructure provided by a third-party provider and shared among customers.
- Private Cloud: Infrastructure dedicated to a single organization.
- Hybrid Cloud: Combines private/on-premises infrastructure with public cloud services.
- Multi-Cloud: Uses services from multiple cloud providers to meet different business/technical needs.

6. Cloud Service Models

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- IaaS: Virtualized compute, storage, and networking resources.
- PaaS: A managed platform for developing and deploying applications.
- SaaS: Ready-to-use software applications delivered over the internet.
- FaaS: Execution of individual functions without managing underlying servers.

Learning Outcomes: Learners can define cloud computing, distinguish deployment and service models, and articulate business benefits and risks.

Activity: Compare a traditional on-premise IT setup vs. a cloud-based setup (case discussion).

Module 2: Virtualization and Cloud Infrastructure

Estimated Duration: 6–7 hours

This module explains the infrastructure technologies behind cloud computing — virtualization, VMs, containers, storage, availability, disaster recovery, and serverless computing.

1. Introduction to Virtualization Technologies

- VMware: Enterprise virtualization technology for creating and managing VMs.
- Hyper-V: Microsoft's virtualization platform for running virtual machines.
- KVM: Linux-based virtualization using the Linux kernel for VM capabilities.

2. Virtual Machines and Virtualization Concepts Explains how a physical server hosts multiple isolated VMs — virtual CPUs, memory, storage, networking, hypervisors, and VM lifecycle management.

3. Virtual Machines vs. Containers

- VMs: Virtualize the hardware and typically run a complete guest OS.
- Containers: Share the host OS kernel while isolating applications and dependencies. Compared on architecture, isolation, resource usage, startup time, and portability.

4. Container Technologies

- Docker: Build, package, distribute, and run applications in containers.
- Kubernetes: Orchestrate and manage containers across multiple machines.

5. Cloud Storage Types

- Block Storage: Storage volumes attached to compute instances.

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- Object Storage: Stores data as objects — used for files, backups, media, large datasets.
- File Storage: Shared file-system-based storage accessible by multiple systems.

6. Load Balancing Distributing incoming traffic across multiple servers/instances to improve availability, scalability, and performance.

7. High Availability Architectures designed to stay available despite individual component failures — redundancy, multiple instances, distributed deployment.

8. Disaster Recovery in Cloud Strategies for recovering applications/data after failures — backup, recovery planning, redundancy, recovery objectives, cloud-based DR approaches.

9. Serverless Computing Concepts Cloud providers manage the underlying infrastructure while developers focus on application logic — event-driven execution, usage-based consumption.

Learning Outcomes: Learners understand how virtualization underpins cloud infrastructure and can differentiate VMs from containers.

Lab: Spin up a virtual machine and a Docker container; compare resource usage.

Module 3: Cloud Service Providers Overview

Estimated Duration: 10–12 hours

Amazon Web Services (AWS)

1. Introduction to AWS — Compute, storage, networking, database, security, monitoring, and application services.

2. Amazon EC2 — Elastic Compute Cloud for creating/managing virtual servers: instance types, AMIs, storage, security groups, SSH access, basic server deployment.

3. Amazon S3 — Object storage: buckets, objects, access control, storage organization, versioning, common use cases.

4. Amazon RDS — Managed relational databases: provisioning, configuration, connectivity, backups, managed operations.

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5. AWS Lambda — Serverless function execution triggered by events, without directly managing servers.

6. Amazon CloudFront — Content delivery service: caching, edge locations, improved delivery performance.

7. AWS IAM — Identity and Access Management: users, groups, roles, policies, permissions, least-privilege access.

8. AWS Cloud Monitoring — Metrics, logs, alarms, dashboards, resource monitoring.

9. AWS Security Concepts — Identity protection, access control, network security, encryption, logging, secure configuration.

Microsoft Azure

10. Introduction to Microsoft Azure — Infrastructure, services, management portal, regions, subscriptions, resource organization.

11. Azure Virtual Machines — Creating/managing VMs: configuration, operating systems, networking, storage, access.

12. Azure Blob Storage — Object storage for files, media, backups, and unstructured data.

13. Azure Active Directory / Microsoft Entra ID — Identity and access management: users, groups, authentication, application access, identity-based security.

14. Azure DevOps and CI/CD Pipelines — Source control, work management, automated builds, testing, application deployment.

15. Security and Compliance in Azure — Security controls, identity protection, monitoring, data security, governance, compliance.

Google Cloud Platform (GCP)

16. Introduction to GCP — Infrastructure, resource organization, regions, zones, core services.

17. Google Compute Engine — Virtual machine service: creating, configuring, managing, scaling compute instances.

18. Google Kubernetes Engine (GKE) — Managed Kubernetes service for deploying/managing clusters on GCP.

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19. BigQuery — Data analytics platform for processing and analyzing large datasets.

20. Google Cloud AI/ML Services — Cloud-based AI/ML capabilities for developing, training, deploying, and consuming AI solutions.

21. Identity Management and Security in GCP — Identity and access management, permissions, service accounts, security controls, access policies.

Learning Outcomes: Learners can navigate the core consoles/services of AWS, Azure, and GCP and identify which provider fits which use case. **Lab:** Deploy a basic VM and storage bucket on each of the three platforms (free-tier accounts).

Module 4: Cloud Networking

Estimated Duration: 6 hours

This module provides the networking foundation required to design secure and scalable cloud infrastructure.

1. Virtual Private Cloud (VPC) — Isolated virtual networks within cloud platforms: network boundaries, subnets, routing, connectivity.

2. Subnetting — Dividing cloud networks into smaller segments: public/private subnets, IP allocation.

3. IP Addressing in Cloud Environments — Public and private IPs and how they're assigned to cloud resources.

4. DNS — Translating domain names into IP addresses to connect users to cloud-hosted applications.

5. Cloud Load Balancers — Distributing traffic across multiple application instances for availability and scalability.

6. Auto Scaling Groups — Automatically adding or removing compute instances based on workload.

7. Content Delivery Networks (CDN) — Caching and delivering content from geographically distributed edge locations to reduce latency.

8. Cloud Firewalls — Network-level security controls regulating inbound/outbound traffic via defined rules.

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9. Security Groups — Instance/resource-level virtual firewall controls to allow or restrict traffic.

10. Cloud Network Architecture — Combining VPCs, subnets, routing, gateways, firewalls, load balancers, and application servers into practical designs.

Learning Outcomes: Learners can design a secure VPC architecture with proper subnetting and traffic control.

Lab: Build a VPC with public/private subnets and configure a security group.

Module 5: Cloud Security and Compliance

Estimated Duration: 7 hours

This module focuses on protecting cloud infrastructure, applications, identities, and data.

1. Cloud Security Best Practices — Least privilege, secure configuration, network segmentation, monitoring, encryption, regular security reviews.

2. Identity and Access Management Across Platforms — Users, roles, groups, policies, and access controls across AWS, Azure, GCP.

3. Authentication and Authorization — Verifying identity vs. determining permitted resource access.

4. Encryption for Secure Cloud Storage — Protecting data at rest and in transit; encryption keys and secure handling.

5. Data Protection — Access control, encryption, backups, monitoring, secure storage practices.

6. DDoS Attacks — Distributed Denial-of-Service attacks and basic mitigation/protection strategies.

7. Malware — Malicious software threats to cloud workloads and basic prevention/detection practices.

8. Data Breaches — Unauthorized access exposure and preventive controls (IAM, encryption, monitoring, secure configuration).

9. Compliance Standards

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- GDPR: Data protection and privacy requirements.
- HIPAA: Protection of healthcare-related information.
- ISO 27001: Information security management framework.
- PCI-DSS: Security requirements for payment card information.

10. Cloud Auditing and Logging — Recording and reviewing cloud activities for security, compliance, troubleshooting, and operations.

11. Security Monitoring Tools — Cloud-native and third-party tools for monitoring security events, logs, and suspicious activity.

Learning Outcomes: Learners can apply IAM policies, encryption, and compliance frameworks to protect cloud workloads.

Activity: Audit a sample cloud environment configuration for security gaps.

Module 6: Cloud DevOps and Automation

Estimated Duration: 8 hours

This module connects cloud infrastructure with modern software development and deployment practices.

1. Infrastructure as Code (IaC) — Defining and managing infrastructure via configuration files instead of manual creation.

2. Terraform — Provisioning and managing cloud infrastructure using declarative configuration.

3. Ansible — Automating server configuration, application setup, and repetitive admin tasks.

4. AWS CloudFormation — AWS's IaC service for defining and provisioning resources via templates.

5. Continuous Integration (CI) — Frequently integrating code changes with automated build and test.

6. Continuous Deployment (CD) — Automated delivery of tested changes into target environments.

7. CI/CD Pipelines in Cloud — Connecting source code, builds, testing, containerization, provisioning, and deployment into an automated pipeline.

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8. Docker — Containerization and application packaging.

9. Kubernetes — Container orchestration: deployments, scaling, and services.

10. Serverless Architecture — Applications built around managed services and functions, with infrastructure managed by the provider.

11. Event-Driven Computing — Applications responding to events such as uploads, API requests, database changes, or schedules.

12. Cloud Monitoring and Logging

- AWS CloudWatch: Monitoring, metrics, logs, alarms, dashboards.
- Azure Monitor: Monitoring Azure resources and applications.
- Google Cloud Monitoring: Monitoring infrastructure, applications, metrics, operational health.

Learning Outcomes: Learners can write basic IaC scripts and set up a CI/CD pipeline for automated deployment.

Lab: Provision infrastructure using Terraform and deploy via a CI/CD pipeline.

Module 7: Multi-Cloud and Hybrid Cloud Strategies

Estimated Duration: 5 hours

This module explores architectures that combine multiple cloud providers or connect cloud infrastructure with on-premises environments.

1. Multi-Cloud Architecture — Using services from two or more cloud providers.

2. Hybrid Cloud Architecture — Combining on-premises/private infrastructure with public cloud services.

3. Multi-Cloud vs. Hybrid Cloud — Key distinctions between the two approaches.

4. API Integration — How applications and cloud services communicate to exchange data/functionality.

5. Cloud Interoperability — The ability of different platforms, applications, and components to work together.

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6. Cross-Cloud Communication — Securely connecting workloads and services across environments.

7. Workload Management — Deciding where applications, databases, and workloads should run (performance, security, availability, cost).

8. Data Migration Strategies — Methods for moving applications/data between on-premises and cloud, or between cloud providers.

9. Cloud-to-Cloud Integration — Patterns for sharing applications, data, APIs, and services across platforms.

10. Multi-Cloud Cost Optimization — Monitoring and controlling costs across multiple providers.

11. Performance Optimization — Maintaining application performance across distributed cloud environments.

12. Multi-Cloud Management Challenges — Different platforms, security models, monitoring tools, skills gaps, networking complexity, vendor management.

Learning Outcomes: Learners can evaluate when a multi-cloud or hybrid approach is appropriate and plan a migration strategy. **Case Study:** Analyze a company's hybrid cloud migration plan.

Module 8: Cloud Cost Optimization and Performance Management

Estimated Duration: 5 hours

This module teaches learners how to control cloud spending while maintaining application performance and reliability.

1. Pay-as-You-Go — Usage-based pricing where organizations pay for resources consumed.

2. Reserved Instances — Cost savings from committing to compute resources for a defined period.

3. Spot Instances — Discounted compute capacity for interruption-tolerant workloads.

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4. Cloud Cost Management — Tracking, analyzing, forecasting, and controlling cloud expenditure.

5. AWS Cost Explorer — Analyzing AWS usage/spending to identify trends and savings opportunities.

6. Azure Cost Management — Analyzing Azure spending, setting budgets, identifying cost-saving opportunities.

7. Resource Utilization — Identifying underutilized or over-provisioned resources.

8. Compute Resource Optimization — Selecting suitable instance types, scaling resources, removing unnecessary compute.

9. Storage Optimization — Reducing storage costs via appropriate classes, lifecycle policies, and data management.

10. Performance Monitoring — Using metrics, logs, and monitoring systems to identify issues.

11. Performance Tuning — Improving performance through resource, application, network, and database optimization.

12. Resource Scaling — Vertical and horizontal scaling; automatic adjustment based on workload.

13. Capacity Planning — Estimating future resource requirements based on growth and usage patterns.

14. Cost Reduction Strategies — Rightsizing, removing unused resources, storage optimization, scaling, budget controls, pricing model selection.

15. Real-World Cost Optimization Case Studies — Analyzing practical scenarios to identify unnecessary expenditure and recommend cost-effective solutions.

Learning Outcomes: Learners can select cost-efficient pricing models and use native tools to monitor spend. **Activity:** Build a sample cost-optimization plan for a given workload.

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Module 9: Cloud Computing Real-World Applications

Estimated Duration: 5 hours

This module demonstrates how cloud technologies are used to build and operate modern applications across industries.

1. Cloud Computing for AI & Machine Learning — Scalable compute, storage, and managed services for AI/ML workloads.

2. Cloud-Based AI/ML Services — Managed services supporting model development, training, deployment, and inference.

3. Serverless Computing — Real-world applications where cloud functions and managed services handle workloads.

4. Microservices Architecture — Dividing applications into independently deployable services; service communication, scalability, container-based deployment.

5. Cloud-Based Big Data — Storing and processing large volumes of structured/unstructured data.

6. Cloud Data Analytics — Processing, analyzing, and visualizing data to derive insights.

7. Cloud Application Architecture — Combining compute, databases, storage, networking, security, monitoring, and scaling services.

8. Healthcare Applications — Healthcare data management, analytics, digital services, AI-assisted applications.

9. E-Commerce Applications — Scalable architectures for online stores, catalogs, payments, and traffic management.

10. Banking & Finance — Financial analytics, digital banking, transaction processing, security, scalable services.

11. Internet of Things (IoT) — Collecting, processing, storing, and analyzing data from connected devices.

12. Gaming — Game hosting, scalable multiplayer services, content delivery, analytics, player management.

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13. Cloud-Based Application Deployment — Moving an application from development to production using automated deployment.

14. Real-World Cloud Architecture Case Studies — Analyzing how cloud services combine to solve business/technical requirements.

Learning Outcomes: Learners can map cloud capabilities to real industry problems and articulate value propositions. **Activity:** Present a cloud solution design for an industry use case of choice.

Module 10: Cloud Certification Preparation & Career Readiness

Estimated Duration: 6–8 hours

This module focuses on certification preparation and helping learners transition into cloud-related careers.

1. AWS Certification Guidance — AWS certification pathways and required knowledge areas.

2. Azure Certification Guidance — Azure certification pathways and associated role skills.

3. GCP Certification Guidance — Google Cloud certification options and technical coverage.

4. Cloud Certification Roadmaps — Selecting paths based on current skills, career goals, and desired role.

5. Certification Exam Preparation — Key concepts, scenario-based questions, practice exercises, revision strategies.

6. Hands-on Cloud Labs — Creating, configuring, deploying, monitoring, and managing cloud resources.

7. Practical Cloud Exercises — Reinforcing theory through real infrastructure tasks (compute, storage, networking, security, deployment).

8. Capstone Project — Applying course concepts to design and implement a practical cloud-based solution end-to-end.

9. Cloud Project Portfolio Development — Documenting and presenting projects via architecture diagrams, Git repos, and demos.

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10. Resume Building — Creating a professional cloud-focused resume highlighting skills, projects, and certifications.

11. Cloud & DevOps Resume Optimization — Presenting relevant technologies and experience using industry-standard, role-specific keywords.

12. Technical Interview Preparation — Cloud architecture, networking, Linux, security, DevOps, containers, automation, troubleshooting.

13. Cloud Interview Questions — Practicing common cloud-related technical and scenario-based questions.

14. Scenario-Based Interview Preparation — Solving real-world scenarios: outages, scaling, security issues, network problems, cost challenges.

15. Career Path Exploration

- *Cloud Architect:* Designs scalable, secure, and reliable cloud solutions.
- *Cloud Security Engineer:* Implements security controls and protects cloud infrastructure/data.
- *Cloud DevOps Engineer:* Automates infrastructure, CI/CD, deployments, and operations.
- *Cloud Administrator:* Manages cloud resources, users, systems, networking, monitoring, and daily operations.

Learning Outcomes: Learners leave with a certification roadmap, a completed capstone project, and an updated resume. **Capstone Project:** Design, deploy, and document a complete cloud solution (compute, storage, networking, security, and automation) on a chosen provider.

Assessment Methods

- Module quizzes (knowledge checks)
- Hands-on labs (graded on completion/functionality)
- Case study analyses
- Final capstone project and presentation

Tools & Platforms Used

AWS Free Tier, Microsoft Azure Free Account, Google Cloud Free Tier, Docker, Kubernetes, Terraform, GitHub/GitLab CI/CD

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Certification Pathways Aligned With This Course

- AWS Certified Cloud Practitioner / Solutions Architect Associate
- Microsoft Certified: Azure Fundamentals (AZ-900) / Administrator Associate
- Google Associate Cloud Engineer

Key Technologies & Platforms Reference

Category	Technologies
Cloud Platforms	AWS, Microsoft Azure, Google Cloud Platform
Compute	AWS EC2, Azure VM, Google Compute Engine
Storage	Amazon S3, Azure Blob Storage, Google Cloud Storage
Database	Amazon RDS and Cloud Database Services
Networking	VPC, Subnets, DNS, Load Balancers, CDN
Containers	Docker, Kubernetes, GKE
Automation / IaC	Terraform, Ansible, CloudFormation
CI/CD	Git, GitHub, Jenkins, Azure DevOps
Monitoring	CloudWatch, Azure Monitor, Google Cloud Monitoring
Data & AI	BigQuery, Cloud AI/ML Services
Virtualization	VMware, Hyper-V, KVM

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